

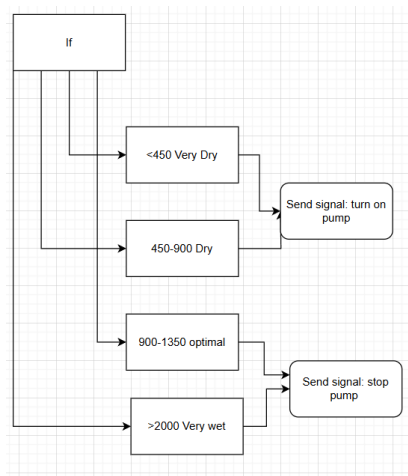
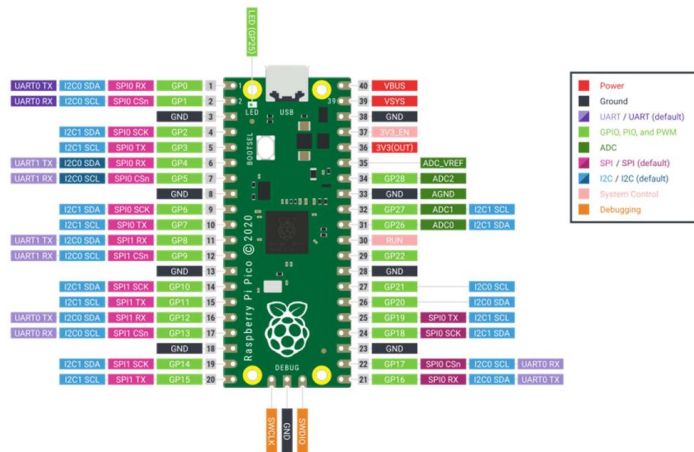
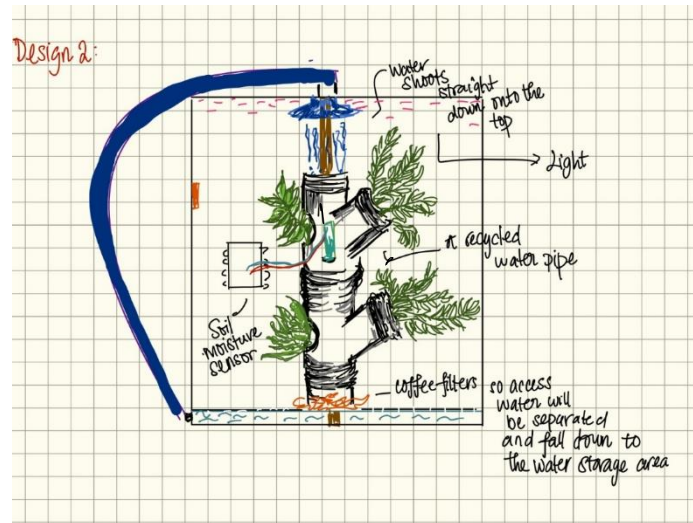
Logbook

February 7th:

Goal:

Finalized plan, created a list of materials, researched spinach, started code

Component	Materials
Showerhead	3D printed
Plant holder	Pipe or water bottles, coffee filters
Box (estimate 4.5x4.5x10in/ 114.3x114.3x254mm)	LEDs, polyethylene (base), 3D printed walls and roof
Electronics housing	Breadboard, wiring, connections for soil and light sensors, <u>Pins:</u> - Soil sensor: 4 and 5 - Grow Light: 15 - Pump: 16 - Photosensor: 28
Water Storage	Pump, tubing



Code flowchart

Plant research:

SPINACH

Light: Full sun to partial shade. At least three to four hours of direct sunlight.

Soil: organically rich, well-draining, moist soil.

Water: frequently but in moderation so soil does not get soggy. Around 139.7 mm of water every week. Best to water multiple times a week

Temperature and humidity: 10-15 degrees Celsius. Do not get it too cold or too hot.

Maintain the soil moisture needs and allow airflow around the plants.

Variables:

soil_m_optimal

soil_m_vdry

moisture

```
soil_m_optimal == True # initialize the soil moisture condition
```

```
soil_m_vdry == False # in the case of soil being very dry
```

```
soil_m_optimal == False # when soil moisture drops below optimal, the condition becomes 'not optimal', or 'dry'
```

```
soil_m_vdry == True # if soil moisture drops very far below optimal, the condition becomes 'very dry'
```

while True:

try:

if soil_m_optimal:

pump.value(0)

```

        elif soil_m_vdry == True:
            #Cycles pump on for 2 seconds and off for 2 seconds
            pump.value(1)
            time.sleep(3)

        else:
            pump.value(0)
            time.sleep(100)

except Exception as e:
    print("ERROR OCCURED:", e)
CODE:

#Grow light imports

from machine import Pin

import time


#Soil sensor imports

from machine import Pin, I2C

from stemma_soil_sensor import StemmaSoilSensor # Save the two Libraries to the Pi Pico
seesaw and stemma_soil_sensor

grow_lights = Pin(15, Pin.OUT) # Grow lights will be controlled with transistor, base of
transistor is connected to GPIO 15


#Water pump imports

from machine import Pin

import time

```

```
#CODE FOR THE GROW LIGHTS
```

```
while True:
```

```
    print("I am here not on soil")
```

```
    # Will cycle grow lights on and off for 4 hour intervals
```

```
    grow_lights.value(1)
```

```
    time.sleep(14400)
```

```
    grow_lights.value(0)
```

```
    time.sleep(72000)
```

```
#CODE FOR THE SOIL SENSOR
```

```
SDA_PIN = 4 # GPIO 4
```

```
SCL_PIN = 5 # GPIO 5
```

```
i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)
```

```
stemma_soil_sensor = StemmaSoilSensor(i2c)
```

```
seesaw = stemma_soil_sensor # Data Sheet = ~200 (very dry) to ~2000 (very wet) 1015 =  
Wet, 331 = Dry
```

```
    # Soil sensor has temp + or - 2 degrees Celsius
```

```
try:
```

```
    moisture = seesaw.get_moisture()
```

```
temperature = seesaw.get_temp()
print(f'Moisture Level: {moisture}, Temperature: {temperature:.1f}{chr(176)}C')
```

```
soil_m_optimal == True
```

```
soil_m_vdry == False
```

```
if moisture <= 950:
```

```
    soil_m_optimal == False
```

```
    if moisture <= 450:
```

```
        soil_m_vdry == True
```

```
except Exception as e:
```

```
    # Sensor reset after 10 sec
```

```
    print(f"Error,Resetting: {e}")
```

```
    time.sleep(10)
```

```
    i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)
```

```
    stemma_soil_sensor = StemmaSoilSensor(i2c)
```

```
    seesaw = stemma_soil_sensor
```

```
#CODE FOR THE WATER PUMP
```

```
    pump = Pin(16, Pin.OUT) # Pump pin is connected to the transistor base, Pump is
connected to 5v of PI Pico with a potentiometer in series to control flow.
```

```
try:
    if soil_m_optimal:
        pump.value(0)
        time.sleep(100)
    elif soil_m_vdry == True:
        #Cycles pump on for 3 seconds and off for 100 seconds
        pump.value(1)
        time.sleep(3)
    else:
        pump.value(0)
        time.sleep(100)

except Exception as e:
    print("ERROR OCCURED:", e)
```

Feb 10: ALT CODE?

Added photoresistor code and altered the grow light code to fit in.

```
from machine import Pin, I2C, ADC
```

```
import time
```

```
from stemma_soil_sensor import StemmaSoilSensor # Save the two Libraries to the Pi Pico  
seesaw and stemma_soil_sensor
```

```
SDA_PIN = 4 # GPIO 4
```

```
SCL_PIN = 5 # GPIO 5
```

```
grow_lights = Pin(15, Pin.OUT) # Grow lights will be controlled with transistor, base of  
transistor is connected to GPIO 15
```

```
pump = Pin(16, Pin.OUT) # Pump pin is connected to the transistor base, Pump is  
connected to 5v of PI Pico with a potentiometer in series to control flow.
```

```
ldr = machine.ADC(28) # Initialize an ADC object for pin 28
```

```
i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)
```

```
stemma_soil_sensor = StemmaSoilSensor(i2c)
```

```
seesaw = stemma_soil_sensor # Data Sheet = ~200 (very dry) to ~2000 (very wet) 1015 =  
Wet, 331 = Dry
```

```
    # Soil sensor has temp + or - 2 degrees Celsius
```

```
ldr_min = 1 # NOTE: will need to initialize later based on photoresistor values
```

```
time_initial = time.time()
```

```
time_on = 0
```

```
grow_lights_on = False
```

```
def soil_moisture():
```

```
    try:
```

```
        moisture = seesaw.get_moisture()
```

```
        temperature = seesaw.get_temp()
```

```
        print(f'Moisture Level: {moisture}, Temperature: {temperature:.1f}{chr(176)}C')
```

```
    soil_m_optimal = True # initialize the soil moisture condition
```

```
    soil_m_vdry = False # in the case of soil being very dry
```

```
    if moisture <= 950:
```

```
        soil_m_optimal = False # when soil moisture drops below optimal, the condition  
        becomes 'not optimal', or 'dry'
```

```
        if moisture <= 450:
```

```
            soil_m_vdry = True # if soil moisture drops very far below optimal, the condition  
            becomes 'very dry'
```

```
    if soil_m_optimal:
```

```
        return 1
```

```
    elif soil_m_vdry:
```

```
        return 2
```

```
    else:
```

```
        return 3
```

except Exception as e:

Sensor reset after 10 sec

print(f"Error,Resetting: {e}")

time.sleep(10)

i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)

stemma_soil_sensor = StemmaSoilSensor(i2c)

seesaw = stemma_soil_sensor

return 0

while True:

ldr_value = ldr.read_u16() # Read the LDR value and convert it to a 16-bit unsigned integer

print("LDR Value:", ldr_value) # Print the LDR value to the console

if (time.time() - time_initial) >= 86400: # checking if a full day, 24 hours, has passed

time_initial = time.time() # resetting both time start markers

time_on = 0

if ldr_value < ldr_min and grow_lights_on == False: # checking if lux is low and the grow lights are off

grow_lights.value(1) # turn the lights on

time_on = time.time() # start timer for how long lights are on, which resets every 24 hours

```
grow_lights_on = True
```

```
print("Light status: dark. Turned lights on.")
```

```
if ((time.time() - time_on) >= 14400) and grow_lights_on == True: # checking if 4 hours of  
lights on have passed in one day, which is the ideal for plant
```

```
    grow_lights.value(0) # turn the lights off
```

```
    grow_lights_on = False
```

```
    print("4 hours of light elapsed. Turned lights off.")
```

```
water = soil_moisture()
```

```
if water == 1: # soil moisture returned as optimal
```

```
    pump.value(0) # pump remains off
```

```
    time.sleep(100)
```

```
    print("Moisture status: optimal.")
```

```
elif water == 2: # soil moisture returned as very dry
```

```
    pump.value(1) # turn the pump on
```

```
    time.sleep(3) # for 3 seconds
```

```
    pump.value(0) # then turn it off again
```

```
    print("Moisture status: very dry. Watered.")
```

```
elif water == 3: # soil moisture returned as dry, but not very dry
```

```
    pump.value(0) # pump remains off
```

```
    time.sleep(100)
```

```
print("Moisture status: dry. Will water later.")
```

```
else:
```

```
    print("Error in soil sensor.")
```

Feb 13

- tested parts of code
- Updated sketch
- discussed material and dimension changes?

```
from machine import Pin, I2C, ADC
```

```
import time
```

```
from stemma_soil_sensor import StemmaSoilSensor # Save the two Libraries to the Pi Pico  
seesaw and stemma_soil_sensor
```

```
SDA_PIN = 4 # GPIO 4
```

```
SCL_PIN = 5 # GPIO 5
```

```
grow_lights = Pin(15, Pin.OUT) # Grow lights will be controlled with transistor, base of  
transistor is connected to GPIO 15
```

```
pump = Pin(16, Pin.OUT) # Pump pin is connected to the transistor base, Pump is  
connected to 5v of PI Pico with a potentiometer in series to control flow.
```

```
ldr = machine.ADC(28) # Initialize an ADC object for pin 28
```

```
i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)
```

```
stemma_soil_sensor = StemmaSoilSensor(i2c)
```

```
seesaw = stemma_soil_sensor # Data Sheet = ~200 (very dry) to ~2000 (very wet) 1015 =  
Wet, 331 = Dry
```

```
    # Soil sensor has temp + or - 2 degrees Celsius
```

```
ldr_max = 50000 # in darker conditions, the ldr value is higher.
```

```
time_initial = time.time()
```

```
time_on = 0
```

```
grow_lights_on = False
```

```
def soil_moisture():
```

```
    try:
```

```
        soil_m_optimal = True # initialize the soil moisture condition
```

```
        soil_m_vdry = False # in the case of soil being very dry
```

```
        if moisture <= 950:
```

```
            soil_m_optimal = False # when soil moisture drops below optimal, the condition  
becomes 'not optimal', or 'dry'
```

```
            if moisture <= 450:
```

```
        soil_m_vdry = True # if soil moisture drops very far below optimal, the condition
becomes 'very dry'
```

```
    if soil_m_optimal:
```

```
        return 1
```

```
    elif soil_m_vdry:
```

```
        return 2
```

```
    else:
```

```
        return 3
```

```
except Exception as e:
```

```
    # Sensor reset after 10 sec
```

```
    print(f"Error,Resetting: {e}")
```

```
    time.sleep(10)
```

```
    i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)
```

```
    stemma_soil_sensor = StemmaSoilSensor(i2c)
```

```
    seesaw = stemma_soil_sensor
```

```
    return 0
```

```
while True:
```

```
    ldr_value = ldr.read_u16() # Read the LDR value and convert it to a 16-bit unsigned
integer
```

```
    print("LDR Value:", ldr_value) # Print the LDR value to the console
```

```

moisture = seesaw.get_moisture()

temperature = seesaw.get_temp()

print(f'Moisture Level: {moisture}, Temperature: {temperature:.1f}{chr(176)}C')

time.sleep(10)


# if (time.time() - time_initial) >= 86400: # checking if a full day, 24 hours, has passed

#time_initial = time.time() # resetting both time start markers

#time_on = 0


if ldr_value > ldr_max and grow_lights_on == False: # checking if lux is low and the grow
lights are off

    grow_lights.value(1) # turn the lights on

    grow_lights_on = True

    print("Light status: dark. Turned lights on.")


if ((time.time() - time_on) >= 10) and grow_lights_on == True: # checking if 4 hours of lights
on have passed in one day, which is the ideal for plant

    grow_lights.value(0) # turn the lights off

    grow_lights_on = False

    print("4 hours of light elapsed. Turned lights off.")


water = soil_moisture()


if water == 1: # soil moisture returned as optimal

    pump.value(0) # pump remains off

```

```
time.sleep(10)
```

```
print("Moisture status: optimal.")
```

```
elif water == 2: # soil moisture returned as very dry
```

```
    pump.value(1) # turn the pump on
```

```
    time.sleep(3) # for 3 seconds
```

```
    pump.value(0) # then turn it off again
```

```
    print("Moisture status: very dry. Watered.")
```

```
elif water == 3: # soil moisture returned as dry, but not very dry
```

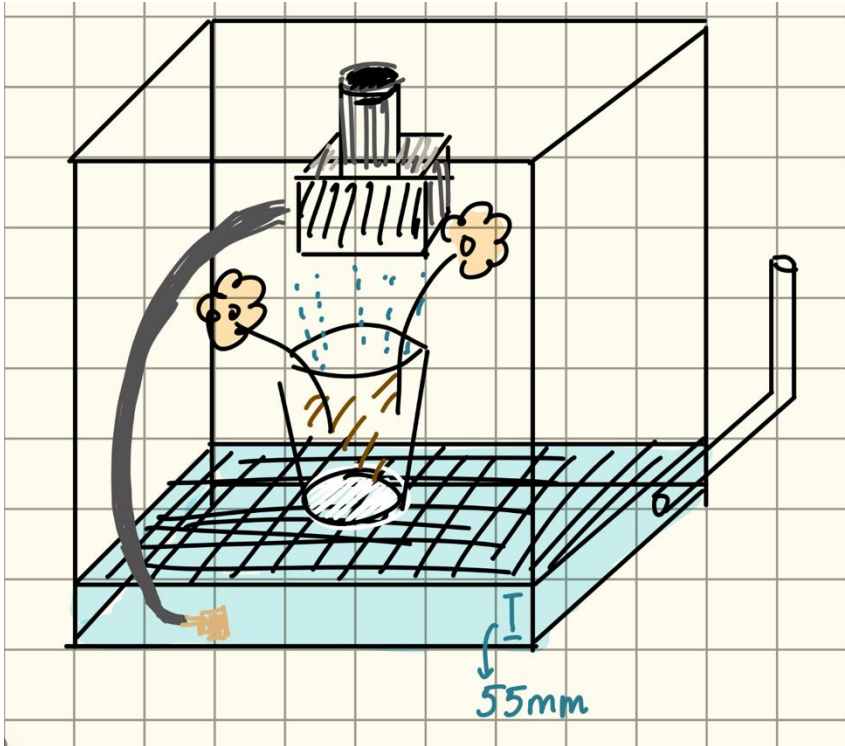
```
    pump.value(0) # pump remains off
```

```
    time.sleep(10)
```

```
    print("Moisture status: dry. Will water later.")
```

```
else:
```

```
    print("Error in soil sensor.")
```



Feb 14:

Prototype:

- Box: 145 mm
- Bottle 225 mm
- Water storage height:
- LDR:

Feb 21:

- Fixed and finalized code, with revised comments (below)
- Put components together
- Worked on presentation and decided which parts to speak on

```
from machine import Pin, I2C, ADC

import time

from stemma_soil_sensor import StemmaSoilSensor


# Pin Definitions

SDA_PIN = 4 # GPIO 4

SCL_PIN = 5 # GPIO 5

grow_lights = Pin(15, Pin.OUT) # Grow lights controlled by a transistor, base connected to
GPIO 15

pump = Pin(16, Pin.OUT) # Pump controlled by a transistor, base connected to GPIO 16

ldr = ADC(28) # Initialize an ADC object for pin 28 (LDR)


i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)

stemma_soil_sensor = StemmaSoilSensor(i2c)

seesaw = stemma_soil_sensor


LDR_MAX = 45000 # threshold for darkness

DAY_DURATION = 86400 # 24 hours in seconds

GROW_LIGHT_DURATION = 14400 # 4 hours in seconds


time_initial = time.time() # start time for the 24-hour cycle

time_on = 0 # time when grow lights were turned on

grow_lights_on = False # track if grow lights are currently on
```

```

def soil_moisture(moisture):

    try:

        soil_m_optimal = True # initialize the soil moisture condition

        soil_m_vdry = False # initialize the very dry condition


        if moisture <= 950:

            soil_m_optimal = False # soil is dry

            if moisture <= 450:

                soil_m_vdry = True # soil is very dry


        if soil_m_optimal:

            return 1

        elif soil_m_vdry:

            return 2

        else:

            return 3


    except Exception as e:

        # sensor reset after 10 seconds

        print(f"Error, Resetting: {e}")

        time.sleep(10)

        i2c = I2C(0, sda=Pin(SDA_PIN), scl=Pin(SCL_PIN), freq=400000)

        stemma_soil_sensor = StemmaSoilSensor(i2c)

        seesaw = stemma_soil_sensor

```

```
return 0
```

```
while True:
```

```
    # read LDR value
```

```
    ldr_value = ldr.read_u16()
```

```
    print("LDR Value:", ldr_value)
```

```
    # read soil moisture and temperature
```

```
    moisture = seesaw.get_moisture()
```

```
    temperature = seesaw.get_temp()
```

```
    print(f'Moisture Level: {moisture}, Temperature: {temperature:.1f}°C')
```

```
    # checking if 24 hours have passed
```

```
    if (time.time() - time_initial) >= DAY_DURATION:
```

```
        time_initial = time.time() # reset the 24-hour timer
```

```
        time_on = 0 # reset the grow light timer
```

```
    if ldr_value > LDR_MAX and not grow_lights_on:
```

```
        grow_lights.value(1) # turn the lights on
```

```
        time_on = time.time() # start the grow light timer
```

```
        grow_lights_on = True
```

```
        print("Light status: dark. Turned lights on.")
```

```
    if (time.time() - time_on) >= GROW_LIGHT_DURATION and grow_lights_on:
```

```
        grow_lights.value(0) # turn the lights off
```

```
        grow_lights_on = False
```

```
print("4 hours of light elapsed. Turned lights off.")

water_status = soil_moisture(moisture)

if water_status == 1: # soil moisture is optimal
    pump.value(0) # pump remains off
    time.sleep(100)
    print("Moisture status: optimal.")

elif water_status == 2: # soil is very dry
    pump.value(1) # turn the pump on
    time.sleep(3) # water for 3 seconds
    pump.value(0) # turn the pump off
    print("Moisture status: very dry. Watered.")

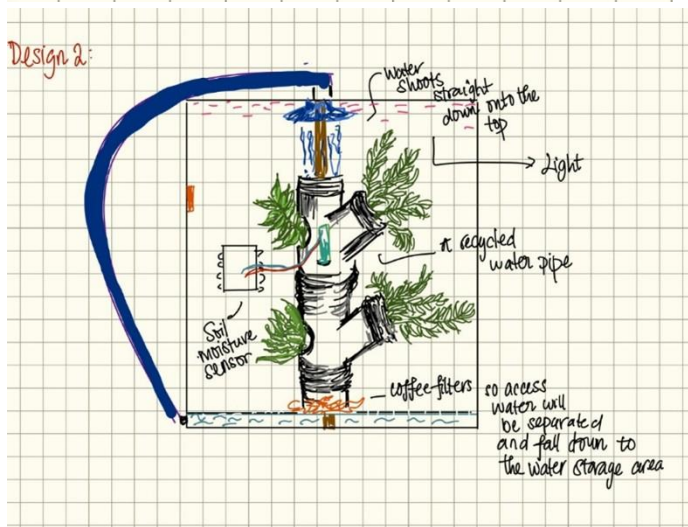
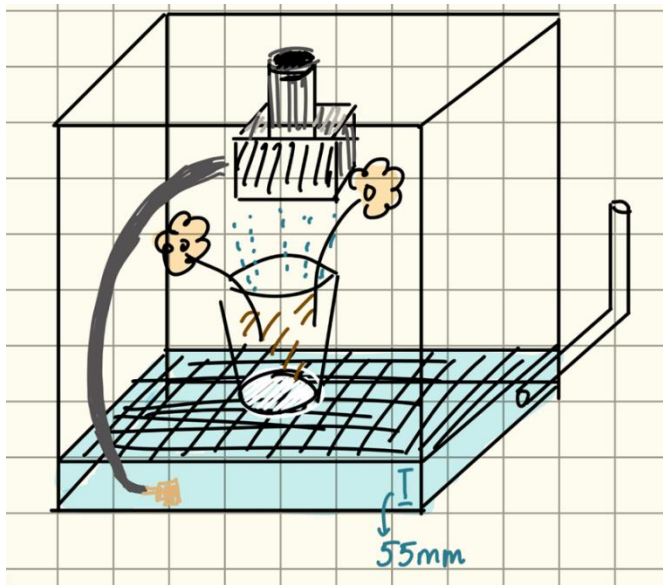
elif water_status == 3: # soil is dry but not very dry
    pump.value(0) # pump remains off
    time.sleep(100)
    print("Moisture status: dry. Will water later.")

else: # error in soil sensor
    print("Error in soil sensor.")

# cpu delay
time.sleep(1)
```

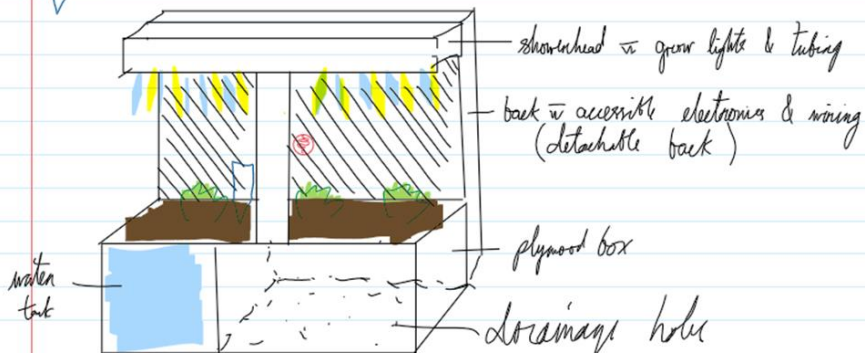
February 25:

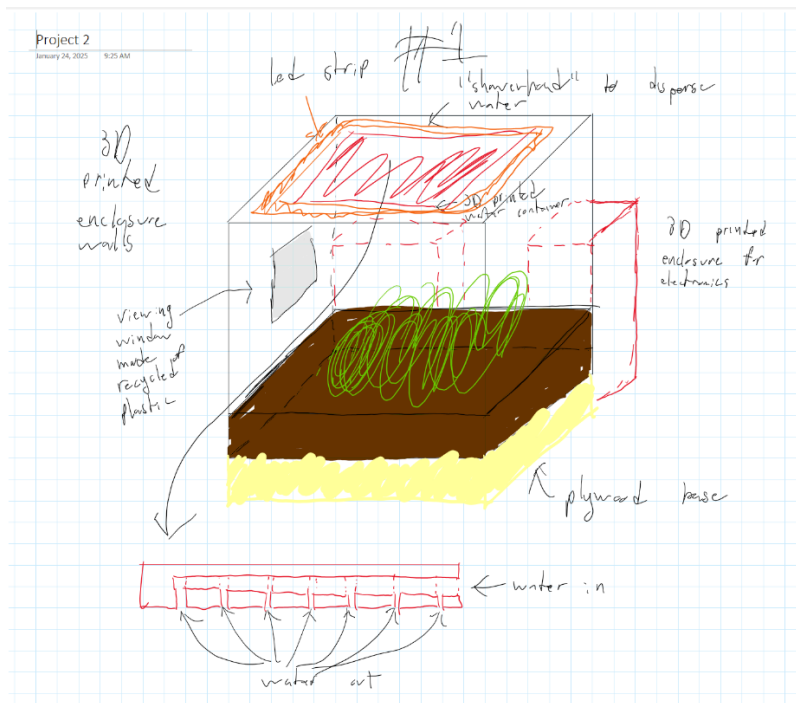
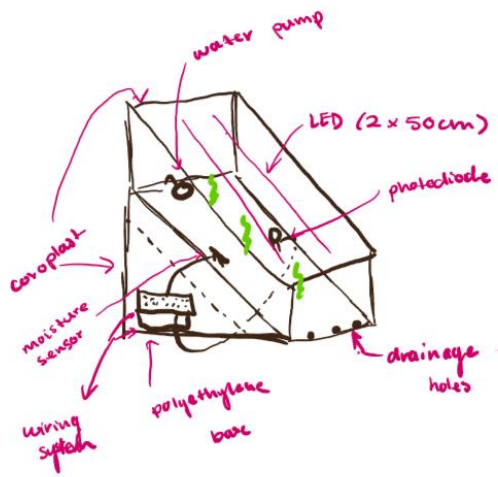
- Put all components together
- Worked on presentation



⊕ - photocurrent (shrinkwrapped?)

▢ - soil moisture sensor





February 26:

- Finished and practiced presentation and slides
- Finished the project

⊗ - photoreactor (shrinkwrapped?)

□ - soil moisture sensor

